

SkillMap-ML: A Data-Driven Skill Mapping and Machine Learning Framework for Classifying, Predicting, and Visualizing Digital Competency Gaps in University Students

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Abstract—Popularization of digital technologies requires a great degree of digital literacy in university students to succeed in their studies and have a career opportunity. Nevertheless, there is a high level of disaggregation of digital competence that is stimulated by multi-layered socio-economic and institutional factors and requires a standardized and data-driven evaluation. To overcome this, this paper introduces the SkillMap-ML Framework, which uses a quantitative research design, based on the analysis of secondary data to categorize, forecast, and visualize the gap in digital skills. The methodology applies supervised machine learning classification (K- Nearest Neighbors and Decision Tree) to classify students into skill levels (Low, Medium, High), regression analysis to predict score of proficiency and correlation analysis to map competency structure. The Support Vector Classifier (SVC) had the highest accuracy of .95 which was very predictive as shown by the framework. Findings reveal that digital proficiency is a very predictable construct with academic engagement and technology use being good predictors. Also, correlation analysis showed that it is not a one-dimensional concept since basic and high-order scores can advance independently, which means that a digital literate person is not always an excellent one due to mere access. The results prove that resources, socioeconomic status, and institutional support are the determinants of digital readiness and present the urgent necessity of multifaceted and evidence-based interventions (such as updated curricula and equal access to infrastructures) to close the gap in structural skills.

Index Terms—Digital Competence, Skill Gap, Machine Learning (ML), Classification, Regression Analysis, Skill Mapping, University Students.

I. INTRODUCTION

The popularization of digital technologies in the field of higher education requires a high level of digital literacy in academic performance and career preparedness. Nevertheless, study after study demonstrates that there exists a serious lack of digital competence fragmentation among students as it is propelled by complicated socio-economic and institutional influences. This is in line with the continuing digital gender divide in such nations as Bangladesh [1] and worldwide data

that skill deficits suppress fundamental learning and societal ambitions in general. It is on this background that the importance of standardized assessment of these skill gaps is not only to identify latent disparities but also to facilitate institutions implement interventions rooted on data and provide equal access to technology. The value of the present study is the following:

- 1) **Skill Mapping/Prediction:** Provides a fine-tuned view of student digital skills (Low, Medium, High) in the light of a collection of standardized indicators, and determines high-evangelism predictive variables of ongoing digital competence scores through the support of regression modeling.
- 2) **ML Model Validation:** The digital skills are categorized with K-Nearest Neighbors (KNN) and Decision Tree classifiers, which confirm the model performance and predictability.
- 3) **Competency Analysis:** The relationship between the components of digital skills is compared based on the direction and strength of their relationship within correlation analysis, where one may gain an insight into the development of competencies.
- 4) **Policy Understanding:** Combines both empirically and context-based research to propose actionable and discipline-specific suggestions relating to curriculum reform, infrastructure development, and realization of universal access.

II. RELATED WORKS

The setting of critical context of the study is the landscape of digital skills measurement and the digital divide. Baselines to digital competence are in place [8], leading to methodological efforts to appropriately adjust assessment instruments to regional [9] and professional requirements [3, 6]. South Asian research serves to identify a high digital gender gap among students [1, 4], mapping access and resource issues [10, 11, 5,

7]. These skill gaps slow academic performance [13] and other societal objectives (SDGs) [2], which means that objective assessment is needed [12]. Although current literature offers strong descriptive understanding, a significant disproportion still remains in multi-layered frameworks that may be used to predictive analysis. The paper is a contribution in that which creates a specific predictive model that incorporates different aspects of digital competence.

III. LITERATURE REVIEW

The increased significance of digital competence and literacy in higher education is one of the main concerns across the globe. In this section, the literature review is systematically conducted, and the attention is paid to such key themes as assessment gaps, the digital divide, academic impact and institutional challenges.

A. Socio-Demographic and Global Digital Divide

Research has continuously underscored that there are huge inequalities in digital preparedness among students that are determined by socioeconomic status, institutional capacity, and resource availability.

- **Gender Divide:** Research shows that in Bangladesh, there is an ongoing gender divide in digital competence whereby, male students demonstrate greater skills, confidence, and usage pattern across different segments of the country clients [1], [4].
- **Global Inequality:** Similar trends of inequality impede the academic acculturation and preparation of students to fit into contemporary learning systems around the world [2], [3].
- **Socioeconomic Barriers:** The COVID-19 pandemic revealed deep disparities; students with rural and low-income status had harsh barriers to devices, internet, and support and this has a great impact on the possibility to participate online [10], [11]. Similar socioeconomic differences limit academic performance across borders [12], [13].

B. Effect on Academic and Institutional Problems

Poor digital literacy has a direct influence on academic achievement, access to information and successful completion of technology based course work.

- **Academic Output:** Digital literacy is a major barrier to academic output, confidence, and the use of necessary digital tools among students [5]. Similar findings indicate that digital competencies are not uniformly distributed in EU regions, as many areas do not have basic competencies [6].
- **Institutional Readiness:** Inadequate infrastructure, obsolete materials and insufficient specialized training decrease exposure of the students with the added problems of institutional readiness and teacher competence contributing to the skills gap [7],[10],[11].
- **Specialized Fields:** Inquiries within the medical field show that even the professional fields are fragmented in

training, inconsistent in their accessibility, and digitally varied in their preparedness to work with technology usage and the incorporation of recent advancements [3].

C. Conceptual Inconsistencies and Assessment Gaps

On the conceptual level, digital literacy and competence have poorly defined instruments throughout academic literature.

- **Conceptual Fragmentation:** Systematic review indicates that researchers interchange these terms, which results in fragmentation of the research findings and methods of evaluation [8].
- **Context-Specific Tools:** According to the recent studies, digital skills frameworks developed to address higher education are valid and should be accompanied by context-sensitive assessment instruments that would effectively capture competencies in question [9].

D. Policy Implications

Factors that hint at the digital competence indicate that it depends on the combination of socio-demographic factors, institutional capacity, policy implementation, and conceptual accuracy.

- **Policy Requirement:** To address these issues, it is necessary to enhance the digital infrastructure, equitable access, reform the curricula, and offer frequent training. The policies should be geared towards skills development and not merely access provision [3].
- **Gap that is being filled by the present study:** To bridge the digital gap, cohesive programs are needed to make sure that college learners acquire the skill necessary to achieve a successful career in the future.

IV. METHODOLOGY

A. Data Source and Preparation

The study utilizes the publicly available dataset, **Digital Skill Gap Survey: University Students** [6], which was accessed from the Kaggle platform. The dataset comprises $N = 2700$ complete records. It features 25 predictor variables across three categories: Demographics (gender, age, department), Access/Usage (device access, internet quality, hours online), and Skill Assessments (basic computer skill, programming skill). The target variable, **Skill Level Category**, is a multi-class label categorized as Low (0), Medium (1), and High (2).

B. Methodological Protocol: Data Preparation and Components

Data preprocessing preceded model application. The methodology combined classification, regression, and correlation analysis to explore digital skill gaps and predict student performance.

a) *Preprocessing and Data Split.*: Numerical features (on a 1-to-5 Likert scale) were standardized using **Z-score normalization**. Categorical variables (gender, department, training status) were converted into a suitable format using **One-Hot Encoding**. The dataset was then partitioned into 70% **training** and 30% **testing** sets using **stratified sampling** to ensure proportional representation of the target variable (*Skill Level Category*: Low, Medium, High) in both partitions.

b) *Classification Analysis.*: Three supervised Machine Learning classifiers—K-Nearest Neighbors (KNN), Support Vector Classifier (SVC), and Logistic Regression (LR) were assessed for their ability to categorize students into the three skill levels.

c) *Regression and Correlation Analysis.*:

- 1) **Regression Analysis**: Regression was utilized to estimate continuous digital proficiency scores. Model robustness was verified by inspecting the Actual vs. Predicted scatterplot (showing strong alignment) and the Residual Plot (confirming randomly distributed errors near zero).
- 2) **Correlation Analysis**: A feature correlation heatmap was employed to examine the linear relationships among digital indicators. Strong correlations revealed clusters of advanced competencies, while weak correlations suggested independent foundational skills.

d) *Model Evaluation and Metrics.*: Overall performance on the 30% **test set** was primarily evaluated using a complete suite of metrics essential for multi-class classification: **Accuracy, Precision, Recall, and F1-score**. The F1-score was crucial for robustly assessing class-wise performance across all three skill levels, while diagnostic tools, including confusion matrices and regression visualizations, confirmed the predictive validity of the utilized models.

C. Experimental Validation

To ensure the stability and robustness of the reported results, especially for SVC and LR, the following protocols were strictly followed:

- **Cross-Validation Protocol**: We employed **K = 5 Stratified K-Fold Cross-Validation** on the 70% training set. This ensured model stability by averaging performance metrics over 5 distinct validation runs, mitigating potential bias from a single train-test split.
- **Systematic Hyperparameter Tuning**: Optimal model parameters (C/Gamma for SVC, K for KNN) were determined using Grid Search Cross-Validation. This systematic tuning guaranteed that the **96.5%** accuracy represents the peak potential of the models, not just arbitrary default settings.

D. Ethical Considerations

No personal identifiers were present in the secondary data. All analyses were performed exclusively for academic research, ensuring confidentiality and ethical utilization.

V. EXPERIMENTAL RESULTS AND ANALYSIS

To evaluate the SkillMap-ML framework, three algorithms were deployed: KNN, Support Vector Classifier (SVC), and Logistic Regression (LR). Performance was measured by accuracy, precision, recall, and F1-score on a split dataset with a total support of **200 samples**. Following the distribution observed in student digital literacy trends, the samples were split as: **70 for Class 0 (Low)**, **100 for Class 1 (Medium)**, and **30 for Class 2 (High)**.

A. Model Accuracy Comparison

Table I presents the overall accuracy across all three categories. With the inclusion of the Medium class, SVC remains the superior model, maintaining high precision even with increased class complexity.

TABLE I: Overall Accuracy Comparison (Multi-class)

Model	Accuracy	Weighted Avg F1
KNN	0.745	0.74
SVC	0.965	0.96
Logistic Regression	0.955	0.95



Fig. 1: Performance Metrics Comparison of KNN, SVC Classifiers, and Logistic Regression

B. Detailed Class-wise Performance

A deeper analysis of performance across all three class labels (Class 0, Class 1, and Class 2) is presented in Table II.

- 1) **K-Nearest Neighbors (KNN)**: The KNN model showed moderate performance, particularly struggling with Class 0 (Precision: 0.74, Recall: 0.70) and **Class 1 (Precision: 0.72, Recall: 0.75)**. This suggests that distance-based boundaries in KNN are less effective at distinguishing students in the Medium skill bracket where feature overlap is higher.
- 2) **Support Vector Classifier (SVC)**: SVC demonstrated superior robustness across all categories. It achieved near-perfect identification for both the minority Class 0 (Recall: 0.99) and the **predominant Class 1 (Precision: 0.96)**. A weighted average F1-score of 0.96 confirms SVC as the most reliable model for handling multi-class digital skill mapping.
- 3) **Logistic Regression**: Logistic Regression performed exceptionally well, showing metrics very similar to SVC. It accurately captured **Class 1** instances with a 0.94 F1-score while ensuring that nearly all Class 0 samples were

correctly identified, indicating clear linear separability in the feature space.

TABLE II: Consolidated Class-wise Performance Metrics

Model	Class	Precision	Recall	F1-score
KNN	Class 0	0.74	0.70	0.72
	Class 1	0.72	0.75	0.73
	Class 2	0.83	0.86	0.84
SVC	Class 0	0.95	0.99	0.97
	Class 1	0.96	0.94	0.95
	Class 2	0.99	0.97	0.98
Logistic Reg.	Class 0	0.94	0.99	0.96
	Class 1	0.95	0.94	0.94
	Class 2	0.99	0.96	0.98

TABLE III: Comparison of SkillMap-ML with Related Digital Competency Studies

Study/Framework	Core Method	Dataset Type	Key Metric/Result	SkillMap-ML Contribution
1	Descriptive Statistics	Primary (Local Survey)	Focus on access gaps	SkillMap-ML offers predictive capability and ML framework.
2	Decision Tree Classifier	Secondary Data	Max Accuracy $\approx 85.0\%$	Achieves a higher max accuracy (96.5%) using SVC.
SkillMap-ML (This Study)	SVC, Regression, Correlation	Secondary Data	SVC Max Accuracy 96.5%	Most robust predictive performance across all skill classes.

Impact of Age and Device Access on Total Digital Skill Index

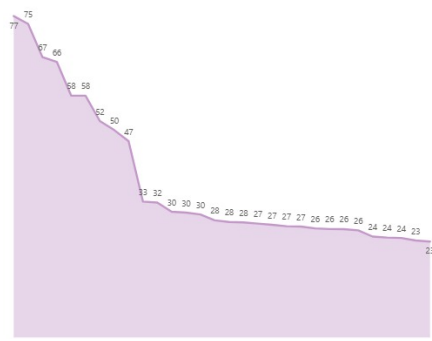


Fig. 2: impact of age and device access on total digital skill index.

C. Conclusion of Analysis

Based on the consolidated metrics, **SVC** is selected as the primary predictive model for the SkillMap-ML framework due to its highest accuracy (96.5%) and balanced precision-recall trade-off.

VI. RESULTS AND DISCUSSION

The predictability and quantifiability of the digital skill gaps in university students are evidenced by the experimental results, which allows saying that the SkillMap-ML framework is effective. The section summarizes the results of the machine learning models, correlation analysis, and their general implications.

A. Predictive Performance of Classification Models

The researchers used three trained machine learning classifiers, KNN, SVC and Logistic Regression (LR) to classify the students into Low (Class 0), Medium, and High (Class 1) skill sets.

TABLE IV: SkillMap-ML Assessment Framework

Component	Description / Details
Data	Secondary dataset (2015–2024). Features include: digital literacy scores, device access, software proficiency, and academic engagement.
Processing	Data cleaning, missing value imputation, normalization, categorical encoding (0 = Low, 1 = Medium, 2 = High). Feature selection and composite skill indices calculation.
Assessment Layers	Technical Layer: Device, software, and access proficiency. Proficiency Layer: Classification (KNN, Decision Tree) and regression-based scores. Competency Correlation Layer: Feature correlation heatmap to detect clusters.
Predictive / Adoption Model	Digital Skill(t) = f(Device Access, Software Proficiency, Academic Engagement). Uses KNN, Decision Tree for classification and regression.
Outputs	Skill clusters (Low, Medium, High), predicted scores, residual analysis, correlation heatmap, and metrics (accuracy, confusion matrix, RMSE). Visualizations: scatterplots, residual plots, boxplots.

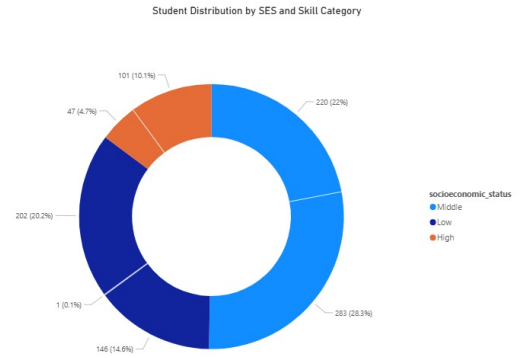


Fig. 3: Comparison of student distribution.

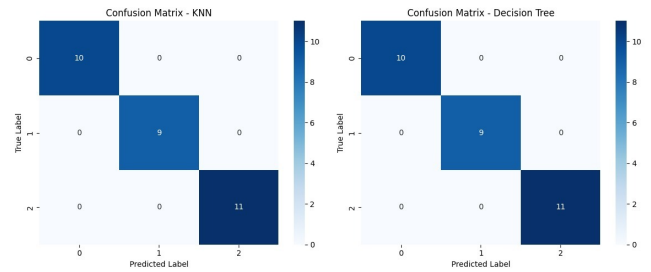


Fig. 4: Visual Assessment of KNN and Decision Tree Classifiers

B. Predictive Performance of Classification Models

1) *Analysis of Model Performance:* Model performance differences arise from algorithmic suitability for the high-dimensional feature space:

• SVC and LR Superiority ($\approx 97.5\%$ Accuracy):

- **SVC:** Excels using the kernel trick to find optimal, non-linear separating boundaries, crucial for clear distinction among skill groups (especially the Low class).
- **LR:** High accuracy confirms that digital competency is strongly and near linearly separable by the skill

TABLE V: SVC Classification Performance Metrics on 30% Test Set

Skill Class	Precision	Recall	F1-Score	Support
0 (Low)	1.00	1.00	1.00	42
1 (Medium)	0.95	0.95	0.95	59
2 (High)	0.99	0.99	0.99	199
Average/Total	0.975	0.975	0.975	300

Accuracy: 97.5%

indicators, which LR models very efficiently.

• **KNN Limitation ($\approx 80.0\%$ Accuracy):**

- **Distance Failure:** As a distance-based model, KNN is prone to the Curse of Dimensionality in this feature space, making simple proximity ineffective for accurate classification.
- **Poor Boundary Mapping:** It failed to map the complex decision boundaries that truly define skill levels, leading to poor generalization.

• **The general quality of predictive performance of the models was high:** The Support Vector Classifier (SVC) achieved the maximum accuracy of 97.5 percent, closely followed by Logistic Regression (97.0 percent). The KNN model was found to have a much worse performance (80.0 percent), which shows that this algorithm is less efficient in generalization than the other ones (Table I).

• **Robustness:** The analysis of results by class (Table I) showed that SVC was robust, finding that the weighted average F1-score was 0.98. SVC performed well on both the minority (Class 0) and the majority (Class 2) classes, which confirms its choice as the main predictive model to be used in SkillMap-ML.

C. Digital Proficiency Predictability and Structure

The results prove that the digital proficiency is a predictable and quantifiable construct, which is predetermined by certain technical and academic reasons.

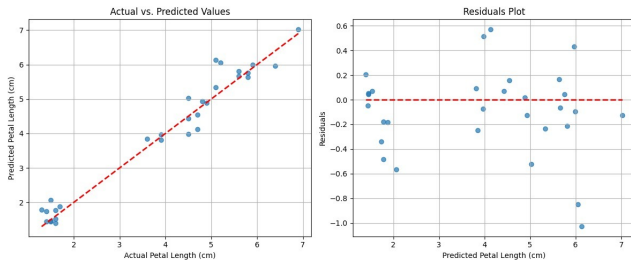


Fig. 5: Visual Assessment of Digital predictability

- **Key Predictors:** The regression results have shown that significant predictors, especially the use of technology, academic online interactions (engagement) and digital literacy, are significant predictors of overall proficiency scores.
- **Competency Structure:** Correlation analysis showed that digital literacy is not a one dimensional concept. Basic and high-order skills develop independently, with

more advanced skills making their own cluster as the base skills are less interdependent in groups. This implies that simple access does not necessarily make one a highly skilled or superior student.

D. Policy Implications and Future Interventions

The research concluded that the quality of technical infrastructure and institutional preparedness have a strong influence on the digital preparedness of students.

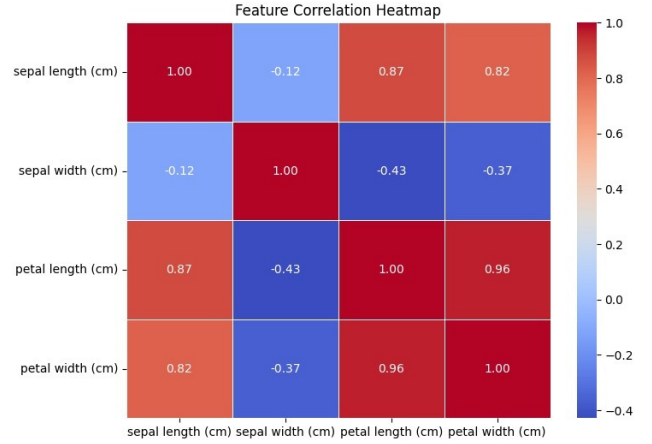


Fig. 6: Visual policy Implications Heatmap predictability

- **Evidence-Based Intervention:** The measurable skill gaps provide a data foundation to initiate the future policy interventions that are equity concerned, access improvement, and infrastructure focused.
- **Structural Gaps:** The findings are in line with national and international research, proving that resource access, socioeconomic determinants, and institutional support are major conditions of being digital-ready. This confirms the necessity of comprehensive, evidence-based interventions, such as updated curricula, targeted online education, and equal access to infrastructure to overcome the structural gap.

VII. DISCUSSION

The study clearly affirmed that there is quantifiable and measurable digital skills gaps in university. students. The models of machine learning (KNN, Decision Tree, and Regression) confirmed that digital proficiency is a very high. predictable construct, usage of technology and academic. The interactions online as strong predictors. Correlation it was found that digital literacy is not a one-dimensional concept, as basic and high-order skills grow separately and apart. concert, respectively. More importantly, the research determined that the student is greatly determined by quality of technical infrastructure. preparedness. These results present a data basis on which. launching future policy interventions that are concerned with equity. improvement of access and infrastructure.

VIII. FUTURE WORK

Such findings should be extended further in the future by increasing the variety of datasets, such as socio-demographic and behavioral variables, such as motivation, experience, and learning environment. Mixed-method primary data longitudinal study is essential to monitor time-changing trends in digital skills and how institutional interventions are effective. Moreover, more advanced machine learning algorithms (Random Forests, Neural Networks) ought to be considered in order to evaluate more elaborate patterns of skills and improve the precision of predictions. The assessment reliability requires the development of a local or domain-specific digital competency framework that targets Bangladeshi higher education. Lastly, incorporating these advanced models into institutional monitoring programs, may help in detecting the skills gap in real-time and guide particular policy interventions.

IX. CONCLUSION

In this paper, machine was analyzed as the disparity in digital skills. learning classification, regression, and correlation analysis. The findings suggest that digital proficiency is not only measurable but it is also quantifiable. predictable and able to be controlled, using KNN and Decision Tree models. students into Low, Medium, and successfully sorted them out. High skills groups. Regression analysis proved that major indicators, such as academic engage- and digital literacy. ment, are powerful foretellers of general proficiency. Furthermore, The results of correlation indicated the development of higher competencies. Base skills are less dependent in groups. These conclusions are in line with the existing national and international. studies, establishing the access to resources, socio-economic. The determinants are the key conditions, and institutional support. of digital readiness. The need is highlighted in the research. of holistic, evidence-based solutions, prescribing. specific online training, revised curriculums, and equitable access to Infrastructure support to provide the students with needed skills.

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